

Sample Paper 9 — Answer Key (Level-2 Difficulty Step-Up)

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Chess** — Hockey, Football, and Cricket are played with a ball; Chess is a board game.
2. **A) Student** — A doctor treats a patient; a teacher teaches a student.
3. **A) IPSTF** — The rule is "+1 letter": $H \rightarrow I$, $O \rightarrow P$, $R \rightarrow S$, $S \rightarrow T$, $E \rightarrow F$.
4. **C) 79** — Two patterns are interleaved: the odd-position terms go 5, 10, 15, 20 (+5 each); the even-position terms go 100, 93, 86, ? (−7 each) — so the next even term is 79.
5. **A) Vikas, Sameer, Tanvi, Umesh, Rhea** — Tanvi is at position 3 (middle). Vikas is at the left end, position 1. Umesh is immediately right of Tanvi, position 4. Sameer sits between Vikas (1) and Tanvi (3), so Sameer is position 2. The only person left, Rhea, takes position 5.
6. **B) 18 m North** — Both North legs add together ($16 + 9 = 25$ m North), and the 7 m South leg is subtracted: $25 - 7 = 18$ m North.
7. **B) Tuesday** — $30 - 11 = 19$ days later; $19 = 2$ weeks + 5 days, so the day of the week moves forward 5 days from Thursday: Tuesday.
8. **D) 45** — 21, 28, and 42 are all multiples of 7; 45 is not.
9. **C) 1,200** — Each term is multiplied by one more than the last multiplier used ($\times 2$, $\times 4$, $\times 5$, $\times 6$): $200 \times 6 = 1,200$.
10. **C) 14th** — Being 18th from the back means 17 people are closer to the back end than Farah. The 4 people who leave from the back end are among those 17, so her position from the back decreases by 4: $18 - 4 = 14$ th.
11. **C) 39** — 13, 29, and 43 are prime numbers; $39 (= 3 \times 13)$ is not.
12. **A) SHQFLO** — The rule is "+3 letters": $P \rightarrow S$, $E \rightarrow H$, $N \rightarrow Q$, $C \rightarrow F$, $I \rightarrow L$, $L \rightarrow O$.
13. **D) Cylinder** — Pentagon, Hexagon, and Octagon are flat (2-D) shapes; a Cylinder is a solid (3-D) shape.
14. **D) Tina** — The finishing order is Sana, then Priya, then Rina, then Tina, so Tina finished last.
15. **B) 18** — The gaps between terms are 1, 2, 3, 4 — each one more than the last. The next gap is 5: $13 + 5 = 18$.

Section 2 — Mathematical Reasoning

16. **A) 8,68,131** — $4,89,675 + 3,78,456 = 8,68,131$.
17. **A) 2,70,325** — $7,60,000 - 4,89,675 = 2,70,325$.

18. **A) 91,602** — $6,543 \times 14 = 91,602$.
19. **A) 5,432** — $97,776 \div 18 = 5,432$ exactly.
20. **C) 36** — $108 = 2^2 \times 3^3$, $144 = 2^4 \times 3^2$. HCF = $2^2 \times 3^2 = 36$.
21. **C) 48** — $16 = 2^4$, $24 = 2^3 \times 3$. LCM = $2^4 \times 3 = 48$.
22. **D) 10/11** — As decimals: $8/9 \approx 0.889$, $5/6 \approx 0.833$, $13/15 \approx 0.867$, $10/11 \approx 0.909$. The greatest is 10/11.
23. **C) 11 km 150 m** — $6250 \text{ m} + 4900 \text{ m} = 11150 \text{ m} = 11 \text{ km } 150 \text{ m}$.
24. **C) 1,700 sq m** — Area = $50 \times 34 = 1,700 \text{ sq m}$.
25. **C) 12 cm** — A regular nonagon has 9 equal sides: $108 \div 9 = 12 \text{ cm}$.
26. **C) 25%** — Profit = $2,250 - 1,800 = ₹450$; profit % = $450/1800 \times 100 = 25\%$.
27. **B) 10%** — If principal P doubles, the interest earned equals P itself. $SI = (P \times R \times T) \div 100 \rightarrow P = (P \times R \times 10) \div 100 \rightarrow R = 100 \div 10 = 10\%$.
28. **C) ₹250** — Total parts = $3+4+5 = 12$; 1 part = $600 \div 12 = 50$; largest share = $5 \times 50 = ₹250$.
29. **A) 9/20** — $0.45 = 45/100$, which simplifies to 9/20.
30. **C) 17 kg 500 g** — $7300 \text{ g} + 5650 \text{ g} + 4550 \text{ g} = 17,500 \text{ g} = 17 \text{ kg } 500 \text{ g}$.
31. **C) 19,800** — $5.5 \text{ hours} \times 3600 \text{ seconds/hour} = 19,800 \text{ seconds}$.
32. **C) 114 cm** — An equilateral triangle has 3 equal sides: $3 \times 38 = 114 \text{ cm}$.
33. **C) 1,980 sq cm** — Area = $55 \times 36 = 1,980 \text{ sq cm}$.
34. **B) 4** — A triangular pyramid (tetrahedron) has 4 triangular faces.
35. **D) 27** — Number of diagonals = $n(n-3)/2 = 9 \times 6/2 = 27$.

Section 3 — Everyday Mathematics

36. **C) ₹45,000** — $3,15,000 \div 7 = 45,000$.
37. **C) 2,520** — Rate = $3,600 \div 10 = 360 \text{ bottles/hour}$; in 7 hours = $360 \times 7 = 2,520$.
38. **C) 30 min** — $6,000 \div 200 = 30 \text{ minutes}$.
39. **C) 4 h 50 min** — Total time 6:45 a.m. to 12:00 noon = 5 h 15 min; subtracting the 25-minute stop gives 4 h 50 min of actual travel.
40. **C) 2 L** — $6500 \text{ mL} + 3500 \text{ mL} = 10,000 \text{ mL}$; $10,000 \div 5 = 2,000 \text{ mL} = 2 \text{ L}$ per bottle.
41. **C) ₹230** — $6 \times ₹42 = ₹252$; $₹252 - ₹22 \text{ discount} = ₹230$.
42. **B) 1,050 L** — $1/6 \times 2,100 = 350 \text{ L}$ used per day; over 3 days = 1,050 L used; $2,100 - 1,050 = 1,050 \text{ L}$ remaining.
43. **C) 72 kg** — $23400 \text{ g} + 19600 \text{ g} + 16250 \text{ g} + 12750 \text{ g} = 72,000 \text{ g} = 72 \text{ kg}$.
44. **C) 56** — $2,352 \div 42 = 56 \text{ rows}$.
45. **C) ₹510** — Sold for $85 \times ₹46 = ₹3,910$; profit = $₹3,910 - ₹3,400 = ₹510$.

Section 4 — Achievers Section (3 marks each)

46. **C) 22** — Working backwards: $(144 + 32) \div 8 = 176 \div 8 = 22$.
47. **C) 40** — Let the first number be x . Then second = $2x$, third = $2x+25$. Sum: $x + 2x + (2x+25) = 225$
 $\rightarrow 5x + 25 = 225 \rightarrow x = 40$.
48. **C) ₹2,250** — After spending $1/3$ on books, $2/3$ of the money remains. After spending $1/5$ of that on stationery, $4/5$ of $2/3 = 8/15$ of the original remains. So $8/15$ of the total = ₹1,200 $\rightarrow$ total = ₹2,250.
49. **C) 239** — Girls = $(2 \times 115) + 9 = 239$. Check: $115 + 239 = 354$. ✓
50. **C) 47** — Each term is the sum of the previous two: $18 + 29 = 47$.
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Scoring guide (out of 60, Level-2-difficulty paper): 32+ is a very strong result at this difficulty. 22–31 is solid. Below 22 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.
